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vivek-bombatkar Application UI Screenshots 52b1f66 · 2 minutes ago

668 lines (465 loc) · 21.2 KB

Preview Code Blame Raw

Multi-Agent AI Assistant with RAG, LLM Routing & Image Generation

Submission Context

This README is prepared as part of the final capstone submission for the **NUS Generative AI Programme**. It documents the design, implementation, and evaluation of a fully integrated multi-agent AI system developed to meet the assignment requirements.

The complete source code for this project is available at the following Git repository:

🔗 https://github.com/vivek-bombatkar/NUS-Capstone-Project

Overview

This project implements a fully integrated multi-agent AI system that combines conversational AI, document intelligence, structured data querying, and generative capabilities into a single application. The system is designed to simulate a real-world AI assistant that can process user queries, route them intelligently, and coordinate multiple specialized agents to generate meaningful outputs.

Business Use Case: AI-Based Customer Feedback Management System

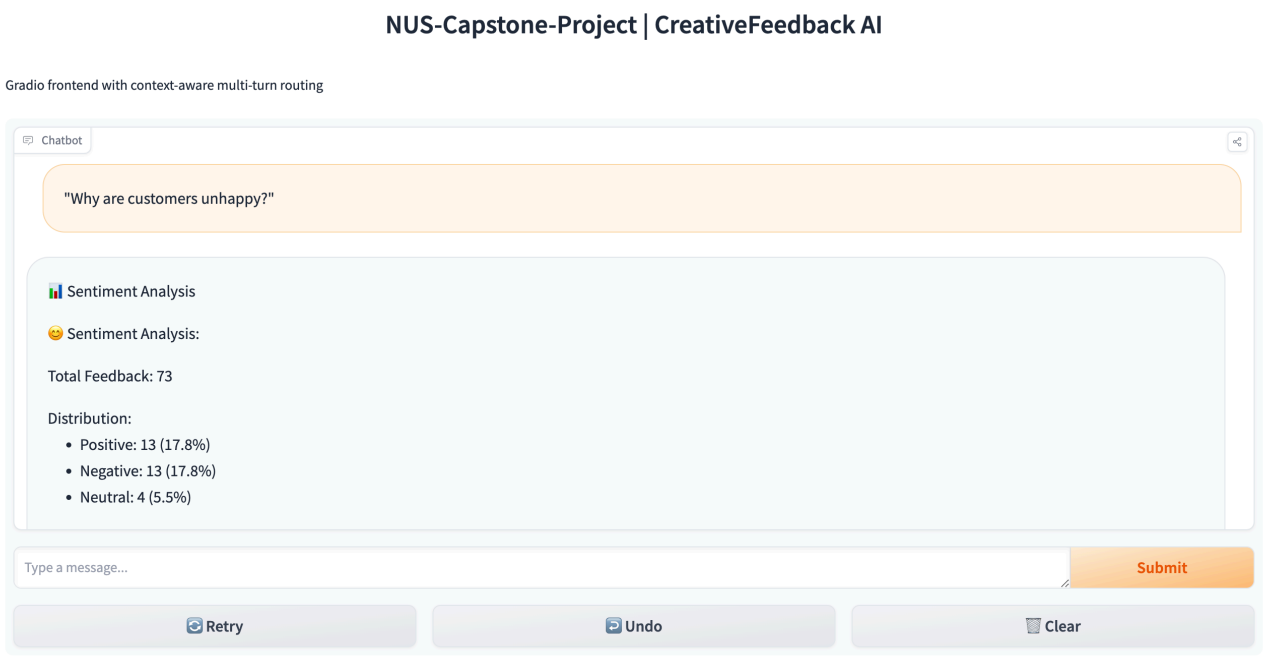
This system helps small and medium-sized enterprises (SMEs) transform large volumes of customer feedback into actionable business insights. Organizations often store feedback across documents, databases, and reports but lack a unified way to extract value from it. This solution demonstrates how AI can centralize feedback analysis, identify sentiment and trends, answer questions from internal documents, and even generate visual representations — enabling better decision-making and faster response to customer needs.

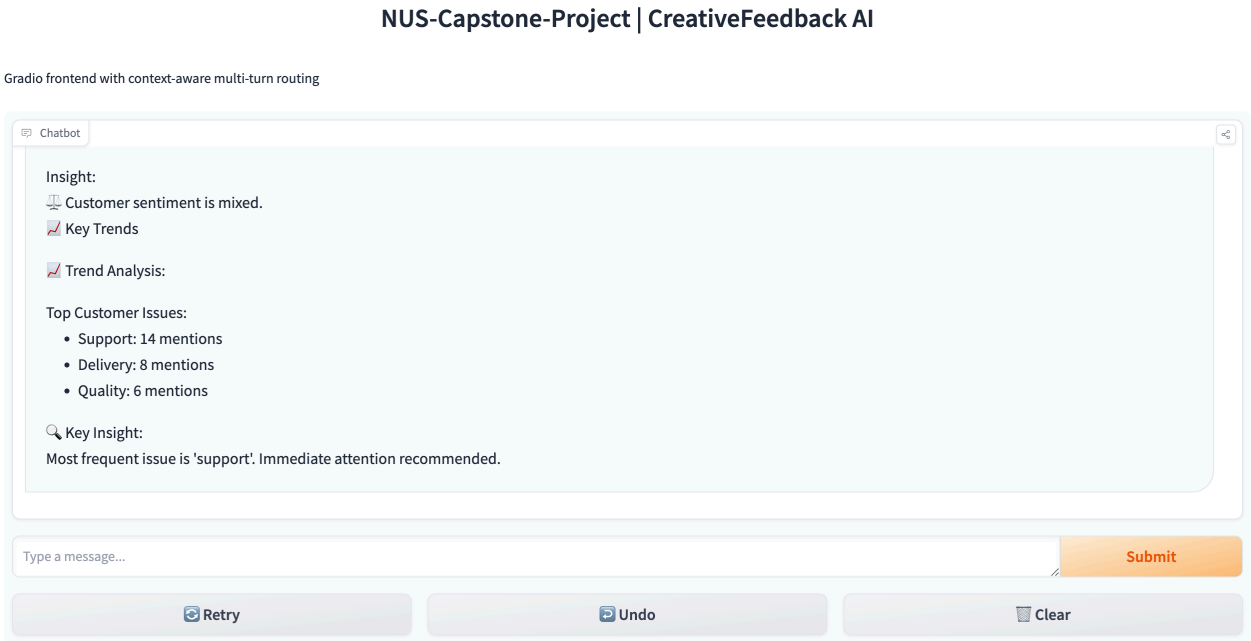
Objectives Achieved

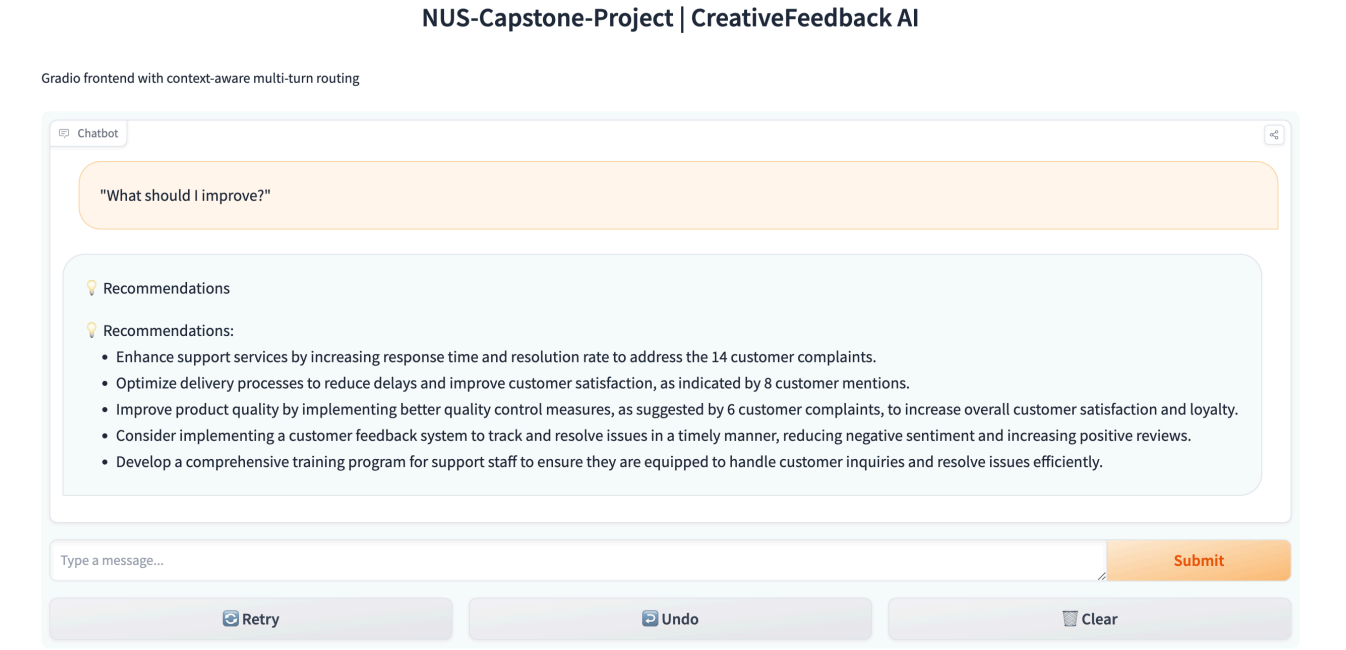
- Conversational interface with limited memory (last 5 exchanges)
- Document-based Question Answering using LLM-based RAG
- Text-to-image generation with prompt engineering
- Multi-agent orchestration through a central controller (Weather, SQL, Recommender)

- LLM-based routing and reasoning
- Fully integrated end-to-end system

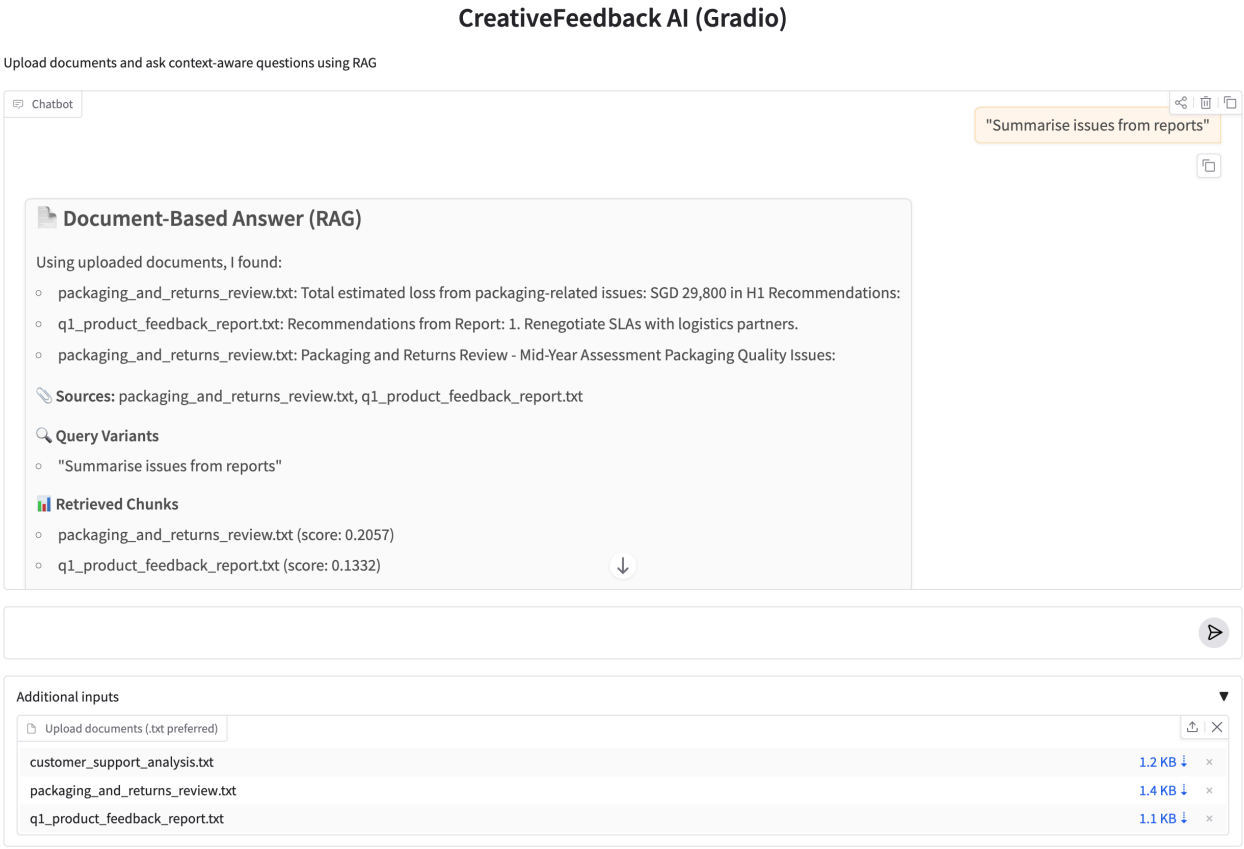
Application UI Screenshots

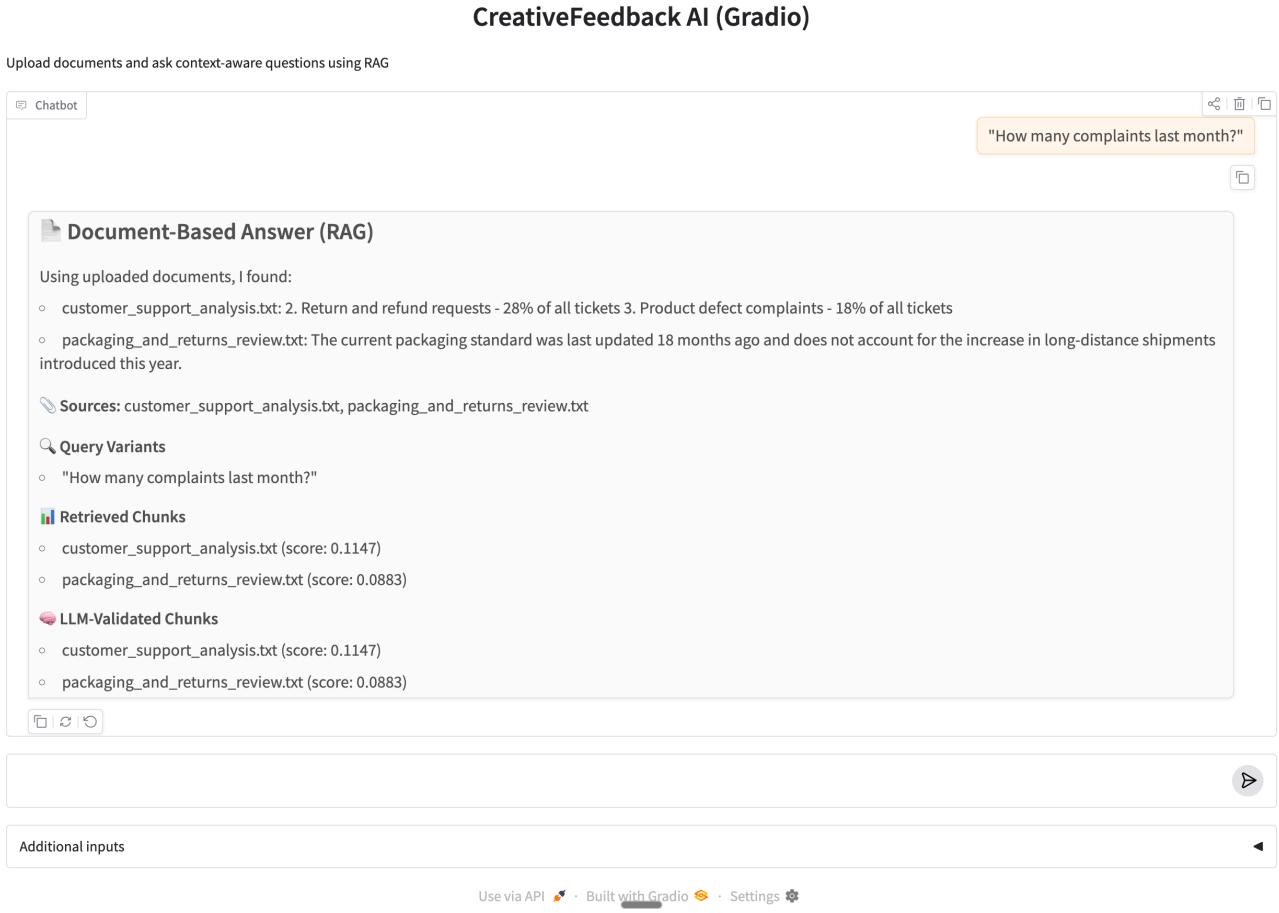


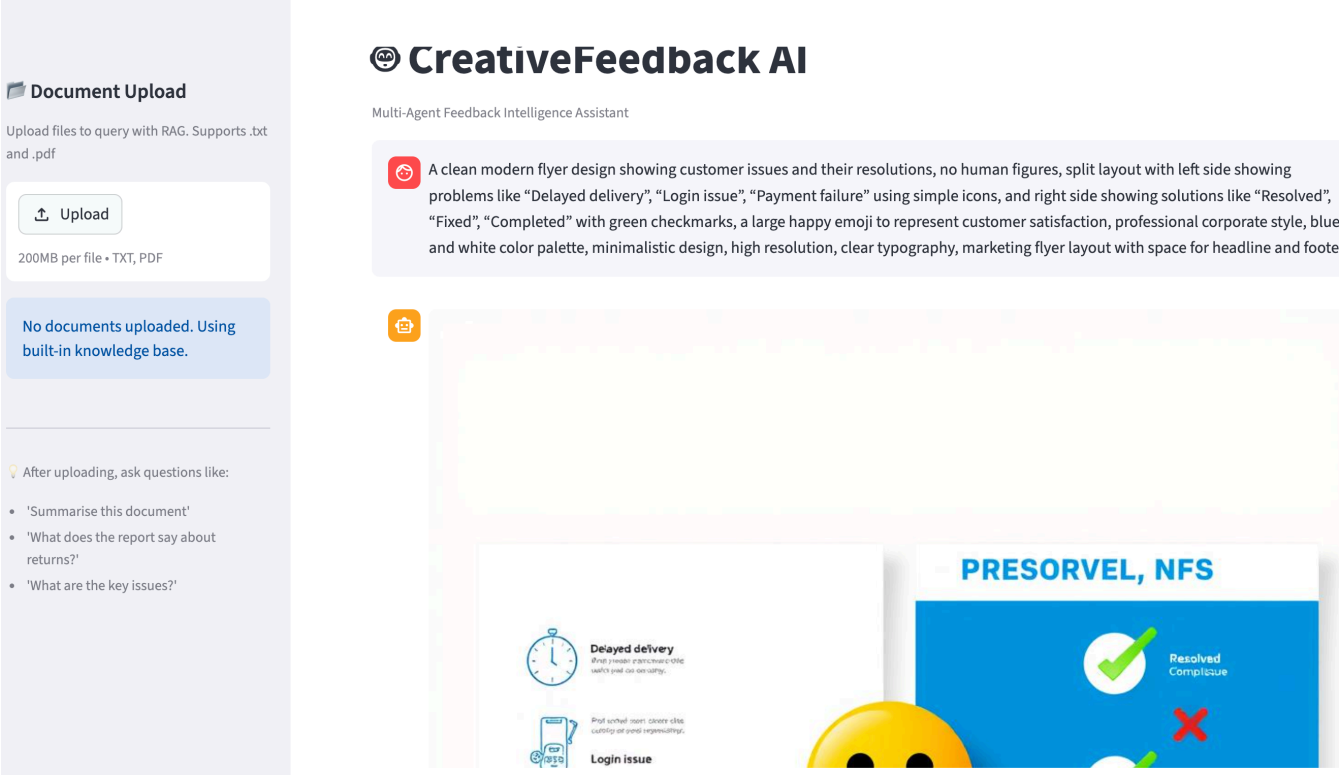



NUS-Capstone-Project | CreativeFeedback AI
Gradio frontend with context-aware multi-turn routing
Chatbot
"What should I improve?"
Recommendations
Recommendations:
Enhance support services by increasing response time and resolution rate to address the 14 customer complaints.
Optimize delivery processes to reduce delays and improve customer satisfaction, as indicated by 8 customer mentions.
Improve product quality by implementing better quality control measures, as suggested by 6 customer complaints, to increase overall customer satisfaction and loyalty.
Consider implementing a customer feedback system to track and resolve issues in a timely manner, reducing negative sentiment and increasing positive reviews.
Develop a comprehensive training program for support staff to ensure they are equipped to handle customer inquiries and resolve issues efficiently.
Type a message...
Submit
Retry
Undo
Clear

CreativeFeedback AI (Gradio)
Upload documents and ask context-aware questions using RAG
Chatbot
"Summarise issues from reports"
Additional inputs
Upload documents (.txt preferred)
customer_support_analysis.txt 1.2 KB
packaging_and_returns_review.txt 1.4 KB
q1_product_feedback_report.txt 1.1 KB
Use via API · Built with Gradio · Settings


CreativeFeedback AI (Gradio)
Upload documents and ask context-aware questions using RAG
Chatbot
"Summarise issues from reports"
Document-Based Answer (RAG)
Using uploaded documents, I found:
packaging_and_returns_review.txt: Total estimated loss from packaging-related issues: SGD 29,800 in H1 Recommendations:
q1_product_feedback_report.txt: Recommendations from Report: 1. Renegotiate SLAs with logistics partners.
packaging_and_returns_review.txt: Packaging and Returns Review - Mid-Year Assessment Packaging Quality Issues:
Sources: packaging_and_returns_review.txt, q1_product_feedback_report.txt
Query Variants
"Summarise issues from reports"
Retrieved Chunks
packaging_and_returns_review.txt (score: 0.2057)
q1_product_feedback_report.txt (score: 0.1332)
Additional inputs
Upload documents (.txt preferred)
customer_support_analysis.txt 1.2 KB
packaging_and_returns_review.txt 1.4 KB
q1_product_feedback_report.txt 1.1 KB


CreativeFeedback AI (Gradio)
Upload documents and ask context-aware questions using RAG
Chatbot
"How many complaints last month?"
Document-Based Answer (RAG)
Using uploaded documents, I found:
customer_support_analysis.txt: 2. Return and refund requests - 28% of all tickets 3. Product defect complaints - 18% of all tickets
packaging_and_returns_review.txt: The current packaging standard was last updated 18 months ago and does not account for the increase in long-distance shipments introduced this year.
Sources: customer_support_analysis.txt, packaging_and_returns_review.txt
Query Variants
"How many complaints last month?"
Retrieved Chunks
customer_support_analysis.txt (score: 0.1147)
packaging_and_returns_review.txt (score: 0.0883)
LLM-Validated Chunks
customer_support_analysis.txt (score: 0.1147)
packaging_and_returns_review.txt (score: 0.0883)
Additional inputs
Use via API · Built with Gradio · Settings



Repository Structure

```
NUS-Capstone-Project/
├── app.py                  # Streamlit UI — main application entry point
├── main.py                 # CLI placeholder (see Running the App below)
├── test_llm.py             # Agent integration tests
├── pyproject.toml          # Dependency manifest (managed via uv)
├── output.png              # Sample generated image (image agent output)
├── agents/
│   ├── sentiment_agent.py  # Analyses feedback sentiment from SQLite
│   ├── trend_agent.py      # Identifies recurring issue patterns
│   ├── recommendation_agent.py  # LLM-based actionable recommendations
│   ├── sql_agent.py        # Structured queries on SQLite
│   ├── rag_agent.py        # Document QA via LLM-based retrieval
│   ├── image_agent.py      # Text-to-image via Hugging Face API
│   └── weather_agent.py    # Live weather lookup via OpenWeatherMap
├── controller/
│   └── router.py           # LLM-based query classifier and agent dispatcher
├── data/
│   ├── feedback.db         # SQLite database with sample customer feedback
│   └── documents/          # Static document store for RAG
└── utils/
    └── db.py               # Database helpers (create_table, insert_feedback)
```

Running the Application

Prerequisites

- Python 3.11+ (see `.python-version`)
- uv package manager

1. Clone the repository

```
git clone https://github.com/vivek-bombatkar/NUS-Capstone-Project.git
cd NUS-Capstone-Project
```

2. Install dependencies

```
uv sync
```

3. Configure API keys

Create a `.env` file in the project root (never commit this file — it is listed in `.gitignore`):

```
export GROQ_API_KEY=your_groq_api_key_here
export HF_API_KEY=your_huggingface_api_key_here
export OPENWEATHER_API_KEY=your_openweathermap_api_key_here
```

API keys can be obtained from:

- Groq: https://console.groq.com
- Hugging Face: https://huggingface.co/settings/tokens
- OpenWeatherMap: https://openweathermap.org/api

4. Launch the app

```
uv run streamlit run app.py
```

The app will open in your browser at `http://localhost:8501`.

Gradio App (Alternative UI)

A lightweight Gradio frontend is available at `gradio_app.py` for quick sharing and Hugging Face Spaces deployment.

Run locally with uv

```
uv run python gradio_app.py
```

What it includes

- Reuses existing `controller.router.handle_query` orchestration
- Supports multi-turn context (last 5 turns)
- Supports document upload + contextual Q&A (RAG over uploaded files)
- Normalizes mixed outputs for chat display:
  - plain text
  - RAG structured payloads
  - `IMAGE_PATH::...` markers

Document-querying (Upload + RAG)

In Gradio, upload one or more documents via the file input and ask questions about them. The app retrieves relevant chunks from uploaded files and answers using only that context.

Tip: Use `.txt` files for best results in this phase.

Deploy to Hugging Face Spaces (Gradio)

1) Create Space

- Go to Hugging Face → **New Space**
- Choose **SDK: Gradio**
- Set visibility/public as needed

2) Push code

Push this repository (or deployment branch) to the Space repository.

3) Configure secrets

In Space settings → **Variables and secrets**, add:

- `GROQ_API_KEY`
- `HF_API_KEY`
- `OPENWEATHER_API_KEY`

4) Launch

Hugging Face will auto-build and run `gradio_app.py` from your repo.

Optional: create share link locally

For local demos, you can temporarily use:

```
# in gradio_app.py
gr.ChatInterface(...).launch(share=True)
```

APIs and Libraries

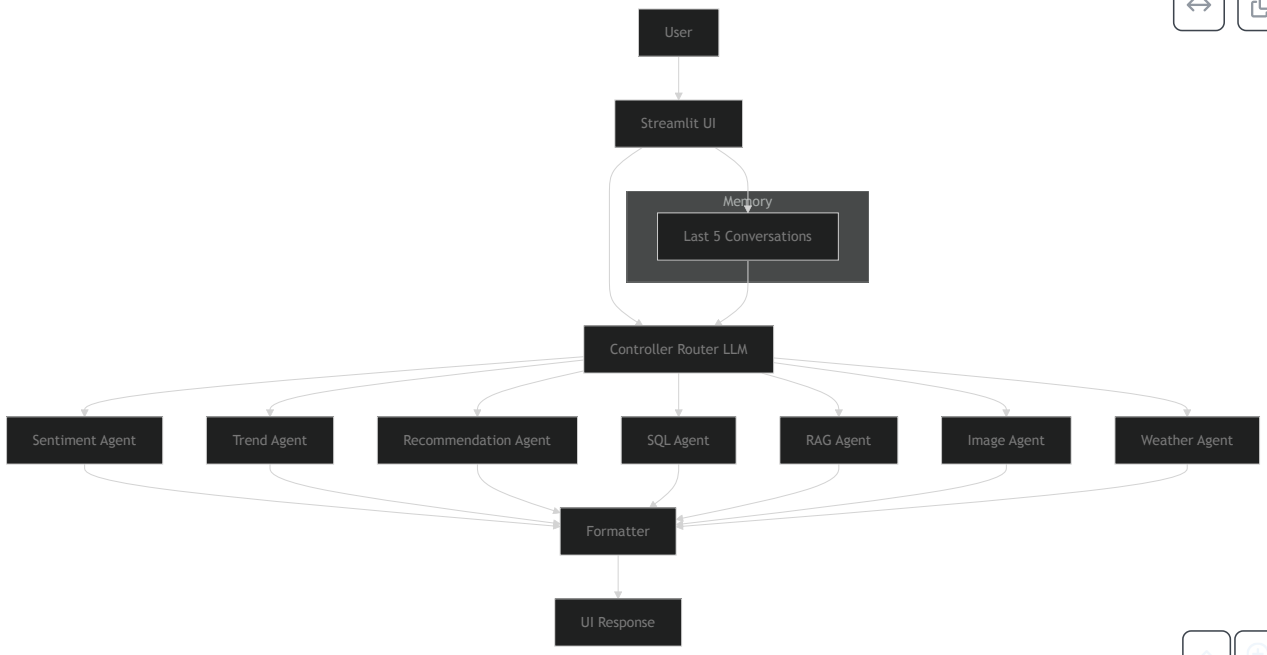
API / Library	Purpose	Authentication	Fallback Behaviour
Groq (LLaMA 3)	LLM for routing, RAG answers, recommendations, prompt engineering	`GROQ_API_KEY` env var	Returns error message to UI
Hugging Face Inference API	Text-to-image generation	`HF_API_KEY` env var	Returns descriptive error; no image rendered
OpenWeatherMap API	Live weather data for Weather agent	`OPENWEATHER_API_KEY` env var	Returns "weather unavailable" message
SQLite (stdlib)	Structured customer feedback storage	None (local file)	N/A
Streamlit	Conversational web UI	None	N/A
LangChain (optional)	Prompt templates	None	Prompts constructed manually

All dependencies are pinned in `pyproject.toml` and locked in `uv.lock`.

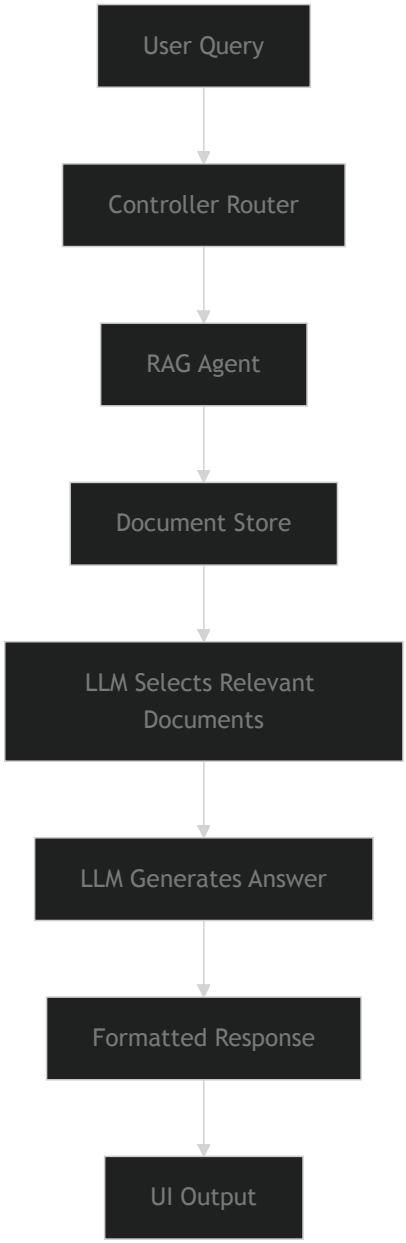
Technical Architecture

Architecture Diagram

The following diagram illustrates the end-to-end flow of the system and how different components interact:



RAG Architecture



Multi-Agent Orchestration Flow

Integration Process

This section describes how individual components were developed and integrated into a single end-to-end system.

Step 1: Independent Agent Development

- Each agent (Sentiment, Trend, Recommendation, SQL, RAG, Image, Weather) was implemented and tested independently
- Focus was on ensuring each agent could handle its specific task reliably

Step 2: Unit Testing of Agents

- A dedicated test script (`test_llm.py`) was used to validate:
  - LLM connectivity
  - API responses
  - SQL query correctness
  - RAG response quality
- This ensured all agents worked in isolation before integration

Step 3: Controller (Router) Implementation

- A central controller (`router.py`) was introduced
- LLM-based routing logic was implemented to:
  - Classify user intent
  - Select appropriate agent(s)
- Output format constraints were enforced for consistent agent invocation

Step 4: Multi-Agent Orchestration

- Agents were connected through the controller
- Dependency flow implemented:
  - Sentiment → Trend → Recommendation
- Outputs from upstream agents were passed as context to downstream agents

Step 5: UI Integration (Streamlit)

- All agent interactions were connected to a single Streamlit interface (`app.py`)
- Controller became the single entry point for all queries
- No manual switching between components

Step 6: Memory Integration

- Session-based memory (last 5 interactions) was added
- Memory is passed to the controller for contextual query understanding

Step 7: End-to-End Testing

- Full system tested using real user queries across:
  - Multi-turn conversations
  - RAG queries
  - SQL queries
  - Image generation
  - Weather queries
- Ensured seamless flow across all components

Step 8: Error Handling and Stability Improvements

- Added `try/except` blocks across agents
- Implemented fallback responses for API failures
- Added logging for debugging integration issues

Core Architectural Layers

1. Presentation Layer (`app.py`)

- Chat-based UI for user interaction
- Maintains session-level conversational memory (last 5 exchanges = `MAX_MEMORY = 5`)
- Memory is passed as context to the controller on every query
- Displays structured text and generated images
- Acts as the entry point for all user queries

2. Controller Layer (`controller/router.py`)

- Implements LLM-based routing instead of rule-based keyword matching
- Uses Groq LLM to classify user intent dynamically
- Outputs a list of agents to be executed
- Supports multi-agent invocation for complex queries

Routing examples:

User Query	Agents Invoked
"Why are customers unhappy?"	Sentiment → Trend
"What should I improve?"	Recommendation
"Summarise feedback reports"	RAG
"How many complaints this month?"	SQL
"Generate a dashboard image"	Image
"What is the weather in Singapore?"	Weather

3. Multi-Agent Layer (`agents/`)

Each agent is independently executable and coordinated through the controller.

Sentiment Agent — Analyses feedback stored in SQLite; computes positive/negative/neutral distribution.

Trend Agent — Identifies recurring issues using keyword aggregation; detects patterns such as delivery delays or product defects.

Recommendation Agent — Consumes outputs from Sentiment and Trend agents; uses LLM reasoning to generate actionable business recommendations. Demonstrates inter-agent context sharing.

SQL Agent — Executes structured queries on SQLite; provides numerical insights such as counts and aggregates.

RAG Agent — Uses a static document store as the knowledge base; LLM selects relevant documents dynamically and generates grounded answers. See RAG Design section below.

Image Agent — Converts user query into an optimised prompt using LLM; calls Hugging Face Inference API for image generation; returns image path for UI rendering.

Weather Agent — Accepts a city name extracted from the user query; calls OpenWeatherMap API; returns current conditions and temperature.

4. Data Layer (`data/`)

- SQLite database (`feedback.db`) for structured customer feedback storage
- Static document repository (`data/documents/`) for RAG knowledge base
- Local file storage for generated images

5. Formatter Layer

- Aggregates outputs from multiple agents
- Structures response into labelled sections: Sentiment Analysis, Trends, Recommendations, Data Insights, Knowledge Insights, Visualisation

Key Implementation Decisions

This section outlines the important architectural and design decisions made while building the system, along with the reasoning behind them.

1. LLM-Based Routing vs Rule-Based Routing

- **Decision:** Use LLM (Groq LLaMA 3) for query classification instead of keyword-based rules
- **Reason:** Enables dynamic understanding of user intent and supports complex, multi-intent queries without hardcoding rules
- **Impact:** Improved flexibility and scalability of the controller

3. SQLite for Structured Data

- **Decision:** Use SQLite for storing customer feedback data
- **Reason:** Lightweight, no external setup required, and sufficient for prototype-scale structured querying
- **Impact:** Easy portability and fast local testing

4. Groq API for LLM Inference

- **Decision:** Use Groq-hosted LLaMA 3 model (`llama3-8b-8192`)
- **Reason:** Faster inference compared to traditional APIs and cost-effective for experimentation
- **Impact:** Improved responsiveness in routing, RAG, and generation tasks

5. Streamlit as Primary UI

- **Decision:** Use Streamlit for the main application interface
- **Reason:** Rapid prototyping, built-in support for chat interfaces, and easy deployment
- **Impact:** Faster development and interactive debugging

6. Modular Multi-Agent Design

- **Decision:** Implement agents as independent modules under `/agents`
- **Reason:** Separation of concerns and easier testing of individual components
- **Impact:** Improved maintainability and scalability

7. Sliding Window Memory (Last 5 Turns)

- **Decision:** Maintain only recent conversation history
- **Reason:** Prevent token overflow and keep context relevant
- **Impact:** Efficient memory usage while preserving conversational continuity

Multi-Agent Coordination

Agents are orchestrated in a dependency-aware pipeline:

```
Sentiment → Trend → Recommendation
```

The Recommendation agent receives the outputs of both prior agents as context, enabling context-aware reasoning rather than isolated execution.

LLM Integration

The system uses Groq-hosted LLaMA 3 models for:

- Query classification (routing)
- Recommendation generation
- Prompt engineering for image generation
- RAG retrieval and answer generation

Model used: `llama3-8b-8192` (configurable)

RAG Design

LLM-Based Retrieval

```
Query → LLM reads document list → LLM selects relevant docs → LLM generates grounded answer
```

What this approach guarantees:

- Simpler architecture with no additional services
- Answers are grounded in the provided document store (not hallucinated)

Image Generation Design

1. User query is passed to an LLM prompt-engineering step
2. LLM generates a detailed, descriptive image prompt
3. Prompt is sent to Hugging Face Inference API (`stabilityai/stable-diffusion-2`)
4. Generated image is saved locally as `output.png` and rendered in the UI

Prompt Experimentation

To improve image generation quality, multiple prompt variations were tested and compared.

1. Basic Prompt (Low Detail)

- **Input:** "Generate a dashboard showing customer satisfaction trends"
- **Output:** Generic chart with minimal structure and poor visual clarity

2. Enhanced Prompt (Moderate Detail)

- **Input:** "A dashboard with bar charts showing customer satisfaction trends over time"
- **Output:** Improved layout with visible charts, but still lacking professional styling

3. Engineered Prompt (High Detail)

- **Input:** "A clean modern data dashboard with bar charts and line graphs showing customer satisfaction scores over 12 months, blue and white colour scheme, professional UI, high resolution"
- **Output:** High-quality, structured, and visually appealing dashboard with clear data representation

Key Learnings

- More descriptive prompts significantly improve output quality
- Including visual style (e.g., colour scheme, layout) enhances realism
- Adding context (e.g., time range, data type) leads to more meaningful images

Conversational Memory

The system maintains a sliding window of the last 5 user/assistant exchange pairs:

```
MAX_MEMORY = 5
# Only the last MAX_MEMORY * 2 messages are kept in session state
```

This context is serialised and passed to the controller on every query, enabling follow-up questions that reference earlier turns. Memory is session-scoped (resets on page reload).

Challenges and Solutions

Challenge	Solution
FAISS / Torch / ChromaDB incompatibility on macOS Intel	Replaced with LLM-based document retrieval
Hugging Face model deprecations and permission errors	Switched to `stabilityai/stable-diffusion-2` ; added fallback error handling
Unicode characters from LLM causing API failures	Applied ASCII sanitisation to all LLM outputs before downstream API calls
Agent coordination lacking context flow	Introduced structured pipeline: Sentiment → Trend → Recommendation with shared context
LLM routing inconsistency on ambiguous queries	Improved system prompt with explicit routing examples and output format constraints

Debugging and Testing

Testing Approach

Each agent was tested in isolation before integration using `test_llm.py` :

```
python test_llm.py
```

The test script covers:

- LLM connectivity (Groq API round-trip)
- SQL agent query execution and result formatting
- RAG agent document selection and answer generation
- Image agent prompt engineering and API response handling
- Controller routing accuracy on sample queries

Integration Testing

End-to-end flow was validated manually via the Streamlit UI with the following test queries:

Test Query	Expected Agent(s)	Verified
"Why are customers unhappy?"	Sentiment + Trend	✅
"What should I improve?"	Recommendation	✅
"Summarise issues from reports"	RAG	✅
"How many complaints last month?"	SQL	✅
"Generate a dashboard image"	Image	✅
"What is the weather in Singapore?"	Weather	✅

Debugging Approach

- Isolated each agent with print-based logging before wiring to controller
- Added API response logging to surface model errors vs. network errors
- Introduced `try/except` with fallback messages at every agent boundary to prevent silent UI failures

Improvements Implemented

- Replaced rule-based routing with LLM-based routing for better generalisation
- Introduced inter-agent context sharing (Sentiment/Trend → Recommendation)
- Improved output formatting with labelled sections per agent
- Added prompt engineering step for image generation quality
- Implemented robust error handling with user-visible fallback messages
- Added ASCII sanitisation to prevent encoding errors in API calls

Known Limitations

- RAG uses LLM-based document selection rather than true vector similarity retrieval
- Memory is session-scoped only; no persistence across sessions
- Image generation latency depends on Hugging Face API queue times (free tier)
- Weather agent requires a valid city name; ambiguous location queries may fail

Future Enhancements

- Integrate vector databases (FAISS or Pinecone) once dependency constraints are resolved
- Improve RAG with document chunking and embedding-based retrieval (e.g., `sklearn` TF-IDF)
- Add persistent memory via a database backend
- Deploy on Hugging Face Spaces or Streamlit Cloud
- Introduce a feedback learning loop where user corrections improve routing accuracy
- Add authentication and multi-user session management

Example Interactions

Customer Insight Analysis

```
User:  Why are customers unhappy?
System: [Sentiment] 62% negative, 28% neutral, 10% positive
        [Trends] Top issues: delivery delays (34%), product defects (21%)
        [Recommendations] Priority actions: ...
```

Document QA

```
User:  Summarise issues from the Q3 report
System: [Knowledge] Based on internal documents: The Q3 report highlights...
```

Image Generation

```
User:  Generate a dashboard showing satisfaction trends
System: [Image] <rendered image>
```

Weather

```
User:  What is the weather in Singapore?
System: [Weather] Singapore: partly cloudy, 31°C, humidity 82%
```

Conclusion

This project demonstrates how multiple AI capabilities can be integrated into a cohesive, production-style prototype. It uses LLMs not just for generation, but for orchestration, reasoning, and decision-making. The system reflects real-world AI architecture patterns and shows how practical constraints (dependency conflicts, API instability) can be addressed through adaptive design choices without compromising the functional requirements.

Final Status

Capability	Status
Conversational interface with memory	✅ Implemented
Document QA (LLM-based RAG)	✅ Implemented
Text-to-image generation	✅ Implemented
Multi-agent orchestration (Weather, SQL, Recommender)	✅ Implemented
LLM-based routing	✅ Implemented
Streamlit UI integration	✅ Implemented
End-to-end executable prototype	✅ Implemented